Viruses are known as obligate intracellular parasites because they cannot independently reproduce outside of a host cell. Viral genomes (RNA or DNA) are enclosed by a protein shell called the capsid. Some viruses, known as enveloped viruses, have an additional phospholipid bilayer that fully envelops the capsid. Viruses that lack this outermost phospholipid covering are labeled non-enveloped. The entry of either enveloped or non-enveloped viruses into a host cell enables the production of viral proteins and the subsequent assembly of viral progeny.

Immune cells (eg, B cells, T cells) protect host organisms against viral infections by recognizing viral particles as foreign. In response to the recognition of foreign particles, antigen-presenting cells ingest these particles and display cleaved antigen fragments on their surface for presentation to other immune cells. The recognition of these displayed antigen fragments then enables the activation of other immune cells to stimulate an immune response, including the production of neutralizing antibodies.

An example of an infectious viral disease is viral hepatitis (ie, inflammation of the liver), which can be caused by any one of five distinct hepatitis viruses (A through E) that infect hepatocytes (liver cells). Hepatitis A, C, and E viruses (ie, HAV, HCV, and HEV, respectively) contain positive-sense, single-stranded RNA (+ssRNA) as genetic material. In contrast, HBV has a partially double-stranded circular DNA genome. HDV is unique because it has a negative-sense, single-stranded RNA (−ssRNA) genome. None of these five viruses are retroviruses.

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Based on the passage, which of the following viruses does NOT have an outer phospholipid bilayer?

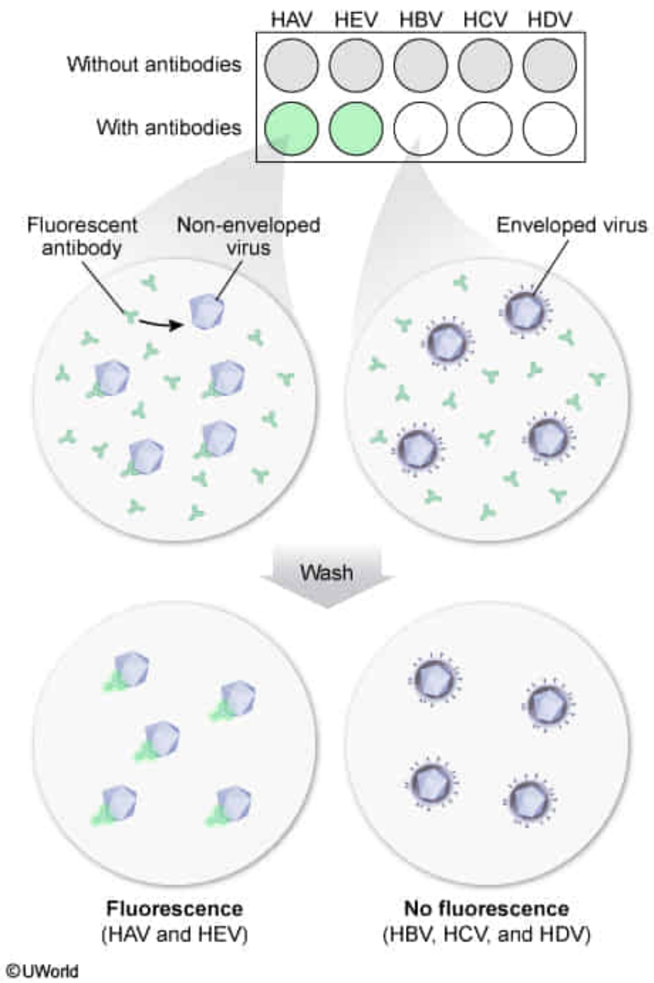
✔ A. HAV [51%]

B. HBV [19%]
C. HCV [3%]
D. HDV [25%]

Correct

51% answered correctly

Explanation:



All viruses contain a protective protein coat called the **capsid**, which is composed of individual capsomere subunits. Viruses that have an additional phospholipid bilayer surrounding the capsid are considered enveloped viruses. Derived from the cell membrane of host cells, the envelope helps the virus evade the immune system and contains envelope proteins required for cellular entry. Enveloped viruses are sensitive to heat, detergents, and changes in moisture that **disrupt the lipid bilayer**. In contrast, **non-enveloped** or naked viruses have only a capsid as a protective outer layer, which is **more resistant** to heat, detergents, and changes in moisture.

In the passage, researchers performed an assay that involved exposing hepatitis viruses to fluorescent antibodies designed to **bind their capsid proteins**. However, bound antibodies were only detected in wells with HAV or HEV. Because the designed antibodies did *not* bind in wells containing HBV, HCV, or HDV, the **capsid proteins of those viruses were likely not exposed** to the environment. The passage states that the capsid proteins of enveloped viruses are fully enclosed within a phospholipid bilayer, which would prevent proteins such as antibodies reaching the capsid enclosed inside. Therefore, HBV, HCV, and HDV are likely **enveloped viruses** whose capsid proteins are enclosed within a phospholipid bilayer **(Choices B, C, and D)**.

In contrast, because wells containing HAV and HEV *did* bind the fluorescent antibodies, their capsid proteins **were likely exposed to the environment** and were able to interact with the antibodies. Therefore, HAV and HEV must be **non-enveloped (naked) viruses** that **do *not*have a phospholipid bilayer**.

Educational objective:
All viruses contain a protective protein coat known as the capsid. Viruses that contain only a capsid as an outer layer are known as non-enveloped or naked viruses and are able to survive in harsh conditions. Viruses with a phospholipid bilayer surrounding the viral capsid are referred to as enveloped viruses; enveloped viruses are more susceptible to changes in environmental conditions.

Time spent:0 sec
Subject:
Biology

QID:401470
System:
Cellular Biology

Viruses are known as obligate intracellular parasites because they cannot independently reproduce outside of a host cell. Viral genomes (RNA or DNA) are enclosed by a protein shell called the capsid. Some viruses, known as enveloped viruses, have an additional phospholipid bilayer that fully envelops the capsid. Viruses that lack this outermost phospholipid covering are labeled non-enveloped. The entry of either enveloped or non-enveloped viruses into a host cell enables the production of viral proteins and the subsequent assembly of viral progeny.

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Researchers perform an experiment in which they infect wild-type mice with either HBV or HEV and subsequently treat them with an antiviral drug that inhibits mouse RNA polymerase activity. Based on the passage, administering this drug to mice infected with either HBV or HEV would most likely result in:

I. Decreased viral load in HBV-infected mice

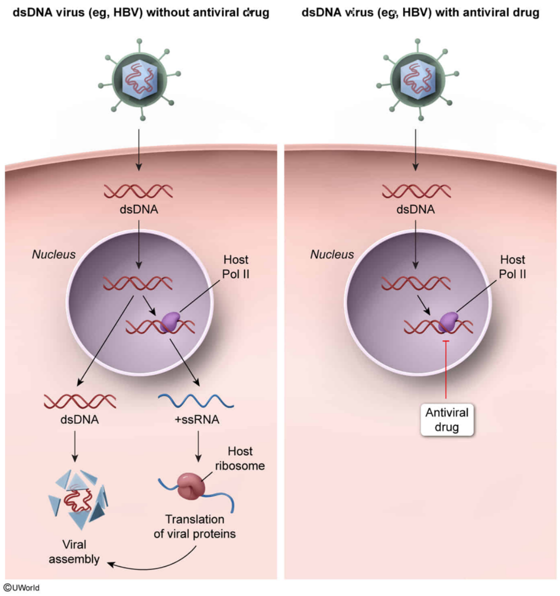
II. Decreased viral load in HEV-infected mice

III. Reduced number of viral fragments displayed by antigen-presenting cells in HBV-infected mice

- ○ A. I and II only [13%]
- ✔ ● B. I and III only [43%]
- ○ C. II and III only [30%]
- ○ D. I, II, and III [12%]

Correct
43% answered correctly

Explanation:



Viral load is measured by the total quantity of viral genetic material inside an organism. Viral genomes consist of either DNA or RNA that is double-stranded (dsDNA) or single-stranded. Once inside a host, viruses rely on cellular (**host**) and/or **viral polymerases** to replicate their genomes and produce mRNAs that can be used for viral protein synthesis.

DNA viruses such as HBV have double-stranded genomes that are **similar** to the host cell's genome; as a result, this type of virus can directly **utilize host enzymes** for transcription and replication. After infecting a cell, the dsDNA genome is trafficked to the nucleus, where host DNA-dependent RNA polymerase (ie, **Pol II**) transcribes the viral genes to mRNA that codes for **viral proteins**, which are subsequently used for viral replication but can also be detected by the immune system.

In this scenario, the administered drug inhibits mouse Pol II. Inhibition of Pol II in mice infected with HBV would prevent transcription of the viral genome, **stopping the production of viral mRNAs and proteins**. Consequently, a **decrease in viral load** is expected as the lack of viral proteins prevents the formation of complete viral particles and viral replication. Similarly, the decrease in viral particles being replicated and released causes **a decrease in HBV fragments displayed** by antigen-presenting cells in HBV-infected mice **(Numbers I and III)**.

(Number II) HEV is a +ssRNA virus, meaning that its genome can *directly serve as an mRNA* molecule in the host cytoplasm. Therefore, HEV can be immediately translated by host ribosomes to form *viral proteins*. One of the viral proteins encoded *must be* the virus's own viral RNA-dependent RNA polymerase, which host cells otherwise lack. Because the administered compounds inhibit only host (ie, mouse) Pol II activity, *viral polymerase activity*

is unaffected. Therefore, viral RNA-dependent RNA polymerase can continue to replicate the HEV genome—first by transcribing the (−) strand complement, then using that as a template to synthesize more of the (+) strand genome—and so the HEV viral load would *not* decrease as a consequence of Pol II inhibition.

Educational objective:
Most double-stranded DNA viruses utilize host machinery and resources to transcribe positive-sense mRNA from viral DNA. Positive-sense ssRNA viruses (except retroviruses) generally replicate their genomes by encoding their own viral RNA-dependent RNA polymerase.

Time spent:0 sec
QID:401471
Subject:
Biology
System:
Cellular Biology

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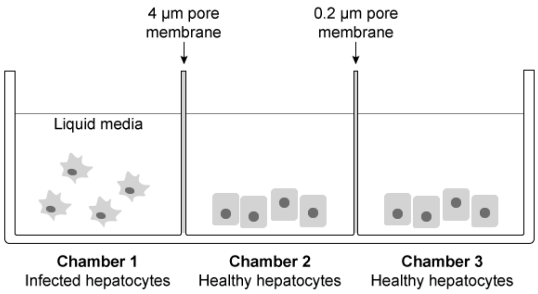
An individual with hepatitis is suspected to have been exposed to either a bacterial strain of *Bartonella henselae* (diameter: 0.5 μm) or HCV (diameter: 0.055 μm). Healthy hepatocytes and hepatocytes infected with serum from the affected patient were grown on chamber slides separated by double membranes, as shown below.

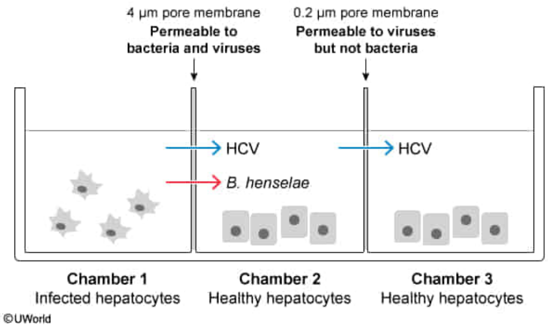


After a 6-hour incubation, both chambers 2 and 3 showed a similar cellular morphology to infected hepatocytes. If the investigators viewed the media from chamber 3 using a conventional light microscope, would they be able to observe the agent that caused the patient's hepatitis?

- A. Yes, because it was caused by HCV [34%]
- B. Yes, because it was caused by *B. henselae* [8%]
- C. No, because it was caused by HCV [50%]
- D. No, because it was caused by *B. henselae* [6%]

Correct
50% answered correctly

Explanation:



The double-membrane system illustrated in the question was used to separate *Bartonella henselae* and HCV on the basis of size. Compared to eukaryotic cells (10–100 µm in diameter), *B. henselae* cells and viruses are much smaller. The average prokaryotic cell is 0.5–2.0 µm in diameter, and most viruses range between 0.02–0.30 µm in diameter. Therefore, both *B. henselae* and HCV can pass through the first membrane from chamber 1 into chamber 2 and infect healthy hepatocytes. In contrast, **movement into chamber 3** is restricted to **HCV** because the 0.2-µm pores are too small for *B. henselae* (0.5 µm) to pass through. Because hepatocytes in **both chambers 2 and 3 were infected** (as shown by their morphology after 6 hours), it can be concluded that **HCV** played a role in the infection and not *B. henselae* **(Choices B and D)**.

The question asks about the use of light microscopy to observe the presence of the causative agent in chamber 3. Viruses are generally too small to be seen by a conventional (ie, nonfluorescent) light microscope. Specifically, HCV is stated to have a diameter of **0.055 µm**. Conventional optical microscopes illuminate samples with visible light, which is then detected by the human eye; therefore, these microscopes are limited to the visible spectrum, with wavelengths going as small as 400 nm. Consequently, the **theoretical limit** of a light microscope (ie, ~half that wavelength) is approximately **200 nm**. Because the **causative agent** passed through the 0.2 µm pore, it is likely the 0.055 µm (ie, 55 nm) HCV particle, which **cannot be seen using a light microscope**.

(Choice A) Although the morphological effects of HCV *on the infected hepatocytes themselves* can be seen by a light microscope, the causative agents (ie, the HCV particles) are too small to visualize by the same method.

Educational objective:

Prokaryotic cells such as bacteria are approximately 10 times smaller than eukaryotic cells. Viruses are about 100 times smaller than bacteria and 1,000 times smaller than eukaryotic cells. Unlike bacteria, viruses cannot be observed with a light microscope because viral particles are smaller than the microscope's maximum resolution.

Time spent:0 sec
Subject:
Biology

QID:401472
System:
Cellular Biology

Viruses are known as obligate intracellular parasites because they cannot independently reproduce outside of a host cell. Viral genomes (RNA or DNA) are enclosed by a protein shell called the capsid. Some viruses, known as enveloped viruses, have an additional phospholipid bilayer that fully envelops the capsid. Viruses that lack this outermost phospholipid covering are labeled non-enveloped. The entry of either enveloped or non-enveloped viruses into a host cell enables the production of viral proteins and the subsequent assembly of viral progeny.

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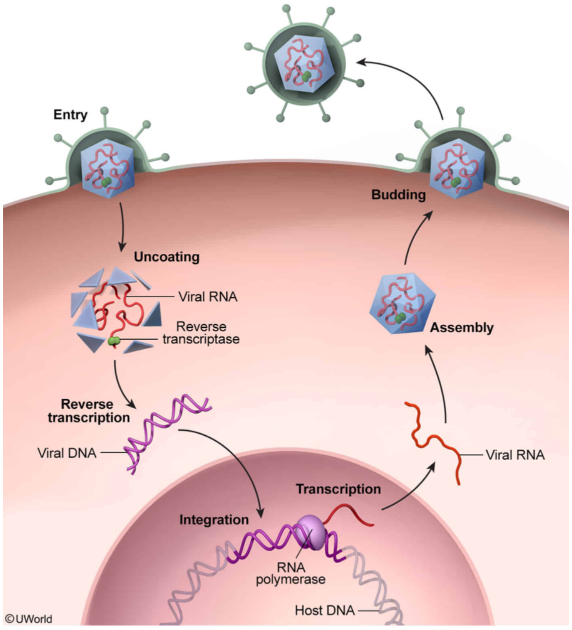
Physicians researching a global hepatitis outbreak discover a new strain of hepatitis (HXV) by analyzing chromosomal DNA of patient's liver cells. They found that HXV has a +ssRNA genome and replicates through DNA intermediates. What characteristic best describes the new hepatitis virus?

- A. It integrates its retroviral +ssRNA into the host genome. [28%]
- ✔ B. It encodes proteins with reverse transcriptase function. [37%]
- C. It synthesizes a complementary RNA strand. [16%]
- D. It translates its own proteins. [17%]

Correct
37% answered correctly

Explanation:

Retrovirus infection



The newly discovered hepatitis strain HXV was identified while analyzing the chromosomal DNA of liver (host) cells. This indicates that unlike the other +ssRNA viruses discussed (ie, HVA, HVC, HVE), the HXV virus is a **retrovirus** because it **integrated with the host genome**. Retroviruses are unique in that they are enveloped and carry two identical **+ssRNA** molecules. Successful viral replication of HXV depends on enzymes that allow its **RNA** viral genome to enter the nucleus and integrate with the host's **DNA** genome.

Retroviruses enter host cells through endocytosis, which allows the capsid and envelope to uncoat (ie, disassemble). Uncoating releases the viral genetic material inside the host cell. RNA viruses such as HXV must then **convert** their genomes from **+ssRNA** into double-stranded DNA (**dsDNA**) before integration into the host genome can occur. The generation of DNA using an RNA template is accomplished using **reverse transcriptase** (ie, RNA-dependent DNA polymerase), which is encoded in the viral genome. Once the dsDNA has been integrated, the viral genome is replicated along with the host cell's own DNA as a lysogenic provirus. Consequently, the original cell containing the integrated viral genome divides and produces descendants that are also infected.

(Choice A) Retroviruses such as HXV can synthesize dsDNA using +ssRNA as a template. It is this dsDNA (*not* +ssRNA) that integrates into the host's DNA genome.

(Choice C) Because the genome of HXV is +ssRNA, the synthesis of a complementary strand would produce a −ssRNA molecule with no usable functions (ie, it neither codes for proteins nor integrates into the host genome). In addition, if the viral ssRNA strands were to hybridize and form a dsRNA molecule, it still could not integrate into the host genome because viral dsRNA must be converted to *dsDNA* for incorporation into the host genome.

(Choice D) According to the passage, viruses are obligate intracellular parasites, meaning they cannot replicate outside a host cell. Independent replication cannot occur because viruses *lack* the ability (ie, resources, enzymes, ribosomes) to produce their own proteins.

Educational objective:

Retroviruses are enveloped, positive-sense, single-stranded RNA viruses that convert their RNA genomes into double-stranded DNA using the enzyme reverse transcriptase. During their lysogenic cycles, retroviruses enter the nucleus and integrate their reversed transcribed DNA with the host genome.

Time spent:0 sec
Subject:
Biology

QID:401473
System:
Cellular Biology

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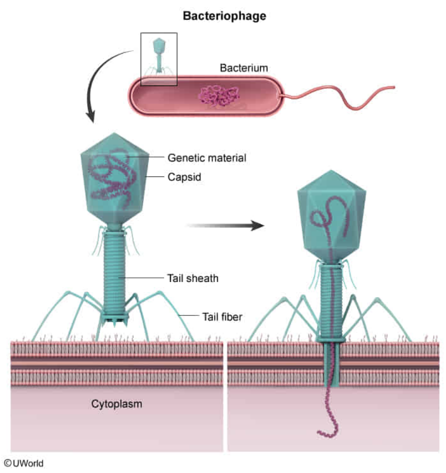
Assume bacteria are infected with a nonliving vector containing HBV genetic material. This vector has been shown to introduce its genomic content into bacterial cells without fully gaining entry into the bacterial cytoplasm. Based on this information, the vector is most likely a:

- A. prion. [7%]
- B. bacterium. [3%]
- C. viroid. [14%]
- ✔ D. bacteriophage. [74%]

Correct
73% answered correctly

Explanation:



Viruses are **nonliving particles** consisting of genetic material wrapped in a protective protein capsid. **Bacteriophages** are viruses that exclusively infect bacteria but do not enter host cells to replicate their genetic material. Instead, they use their **tail sheath**, a structure that injects the phage genome into a bacterium. The remaining viral structures of the phage, such as the **tail fibers**, the capsid, and the tail sheath, are left outside the bacterium.

(Choice A) A prion is a misfolded protein that acts as an infectious agent by inducing other normal proteins to change their secondary structure and become misfolded. These less-soluble, misfolded proteins aggregate and can cause disease. Prions do not contain genetic material and cannot transform bacteria.

(Choice B) Bacteria are living prokaryotic organisms; they do not have tail fibers nor inject *viral* DNA into other bacterial cells.

(Choice C) Viroids are not viruses; instead, they are pathogenic, circular, single-stranded RNA molecules lacking protein coats that affect primarily plants. They typically silence the expression of specific genes and inhibit protein synthesis by binding RNA sequences. Viroids enter cells by hiding inside viruses or through damaged tissue; they *do not* use the mechanisms described in the question.

Educational objective:
Bacteriophages are viruses that exclusively infect bacterial cells. They contain tail fibers that allow them to recognize and attach to the cell membrane, and a tail sheath that injects the viral genome into the bacterium. Viroids and prions are considered subviral particles because they are smaller in size and complexity compared to viruses.

Time spent: 0 sec	**QID:** 401474
Subject: Biology	**System:** Cellular Biology